

Tasks on VPC till NAT Gateway

1. Create a VPC with 2 private and 2 public subnets.

Purpose: To establish an isolated network infrastructure for hosting applications with separation between public-facing and private resources.

STEPS TO CREATE A VPC AND SUBNETS:

VPC:

- Navigate to VPC Dashboard
- Click "Create VPC"
- Select "VPC only"
- Name: firstvpc
- IPv4 CIDR: **192.168.0.0/24**
- IPv6 CIDR: No IPv6 CIDR block
- Tenancy: Default
- Click "Create VPC"

Create VPC info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings info

Resources to create info
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional info
Creates a tag with a key of 'Name' and a value that you specify.

firstvpc

IPv4 CIDR block info
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
192.168.0.0/24
CIDR block size must be between /16 and /28.

IPv6 CIDR block info
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy info
Default

Tags info
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key info
Name

Value - optional info
firstvpc

You can add 49 more tags.

Cancel Preview code **Create VPC**

SUBNET:

Public Subnet 1

- VPC Dashboard > Subnets > Create subnet
- VPC ID: Select firstvpc
- Subnet 1:
- Name: **public-subnet-1**
- Availability Zone: **us-east-1a**

- IPv4 CIDR: **192.168.0.0/26**
- Add new subnet (Subnet 2):
- Name: **public-subnet-2**
- Availability Zone: **us-east-1b**
- IPv4 CIDR: **192.168.0.64/26**
- Click "Create subnet"

Public Subnet 1

The screenshot shows the AWS Management Console interface for creating a new subnet. The breadcrumb navigation indicates the path: VPC > Subnets > Create subnet. The page title is "Subnet settings" with a subtitle "Specify the CIDR blocks and Availability Zone for the subnet." Below this, it indicates "Subnet 1 of 1". The form contains four sections: "Subnet name" with a text input field containing "public-subnet-1"; "Availability Zone" with a dropdown menu showing "United States (N. Virginia) / use1-az2 (us-east-1a)"; "IPv4 VPC CIDR block" with a dropdown menu showing "192.168.0.0/24"; and "IPv4 subnet CIDR block" with a text input field containing "192.168.0.0/26" and a "64 IPs" label. At the bottom of the form, there are navigation arrows: back, forward, up, and down.

Public Subnet 2

The screenshot shows the AWS Management Console interface for creating a new subnet. The breadcrumb navigation indicates the path: VPC > Subnets > Create subnet. The page title is "Subnet settings" with a subtitle "Specify the CIDR blocks and Availability Zone for the subnet." Below this, it indicates "Subnet 1 of 1". The form contains four sections: "Subnet name" with a text input field containing "public-subnet-2"; "Availability Zone" with a dropdown menu showing "United States (N. Virginia) / use1-az4 (us-east-1b)"; "IPv4 VPC CIDR block" with a dropdown menu showing "192.168.0.0/24"; and "IPv4 subnet CIDR block" with a text input field containing "192.168.0.64/26" and a "64 IPs" label. At the bottom of the form, there are navigation arrows: back, forward, up, and down.

Follow the same process for private subnets

Subnet 3:

- Name: private-subnet-1
- Availability Zone: us-east-1c
- IPv4 CIDR: 192.168.0.128/26

Subnet 4:

- Name: private-subnet-2
- Availability Zone: us-east-1d
- IPv4 CIDR: 192.168.0.192/26

Private Subnet 1

The screenshot shows the AWS Management Console interface for creating a new subnet. The breadcrumb navigation indicates the path: VPC > Subnets > Create subnet. The page title is 'Subnet settings' with a subtitle 'Specify the CIDR blocks and Availability Zone for the subnet.' Below this, it indicates 'Subnet 1 of 1'. The 'Subnet name' field is set to 'private-subnet-1'. The 'Availability Zone' is set to 'No preference'. The 'IPv4 VPC CIDR block' is set to '192.168.0.0/24'. The 'IPv4 subnet CIDR block' is set to '192.168.0.128/26', with a note indicating '64 IPs' available. The interface includes a search bar, account ID (7348-4675-3465), and user name (omerfarooq257) in the top right corner.

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 7348-4675-3465 omerfarooq257

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
private-subnet-1
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.0.0/24

IPv4 subnet CIDR block
192.168.0.128/26 64 IPs

Private Subnet 2

The screenshot shows the AWS Management Console interface for creating a new subnet. The breadcrumb navigation indicates the path: VPC > Subnets > Create subnet. The page title is 'Subnet settings' with a subtitle 'Specify the CIDR blocks and Availability Zone for the subnet.' Below this, it indicates 'Subnet 1 of 1'. The 'Subnet name' field is set to 'private-subnet-2'. The 'Availability Zone' is set to 'No preference'. The 'IPv4 VPC CIDR block' is set to '192.168.0.0/24'. The 'IPv4 subnet CIDR block' is set to '192.168.0.192/26', with a note indicating '64 IPs' available. The interface includes a search bar, account ID (7348-4675-3465), and user name (omerfarooq257) in the top right corner.

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 7348-4675-3465 omerfarooq257

VPC > Subnets > Create subnet

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
private-subnet-2
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
No preference

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
192.168.0.0/24

IPv4 subnet CIDR block
192.168.0.192/26 64 IPs

The created VPC can be seen in the image below.

Subnets (4) Info							
Find subnets by attribute or tag							
☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	
☐	public-subnet-1	subnet-047d43ebf78f5aeab	Available	vpc-056e352fea5d9f0d7 firstvpc	Off	192.168.0.0/26	
☐	private-subnet-1	subnet-092b11187c86d689c	Available	vpc-056e352fea5d9f0d7 firstvpc	Off	192.168.0.128/26	
☐	public-subnet-2	subnet-0e495261d7e282845	Available	vpc-056e352fea5d9f0d7 firstvpc	Off	192.168.0.64/26	
☐	private-subnet-2	subnet-00c0bf575be6a7d76	Available	vpc-056e352fea5d9f0d7 firstvpc	Off	192.168.0.192/26	

2. Enable DNS Hostname in VPC

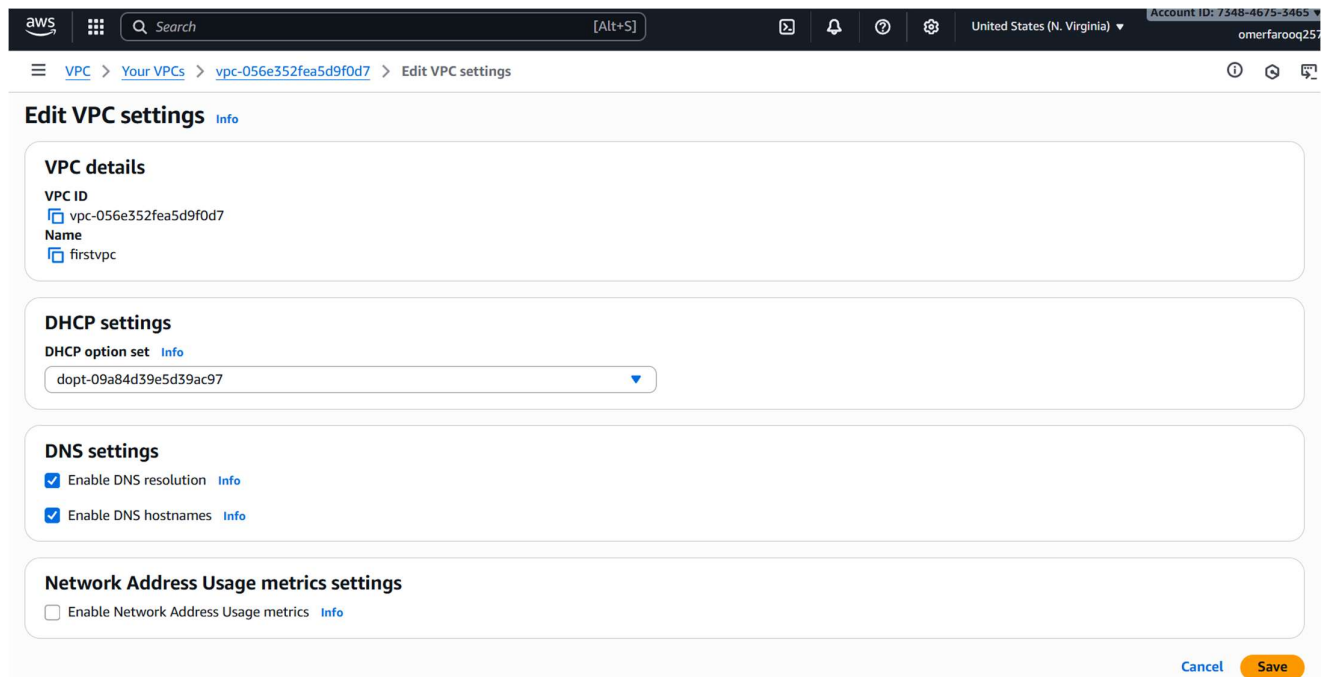
Purpose: To allow instances to receive DNS hostnames and enable hostname-based communication within the VPC.

STEPS TO ENABLE DNS HOSTNAME IN A VPC:

- Navigate to VPC Dashboard
- Select your VPC (firstvpc)

The screenshot shows the AWS VPC console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and account information. The main content area is titled 'Your VPCs' and shows a list of VPCs. The 'firstvpc' VPC is selected, and its details are shown below. A green notification banner at the top indicates a successful modification. The 'Actions' menu is open, showing options like 'Create default VPC', 'Create flow log', 'Edit VPC settings' (which is highlighted), 'Edit CIDRs', 'Manage middlebox routes', 'Manage tags', and 'Delete VPC'.

- Click "Actions" > "Edit VPC settings"
 - Under DNS settings:
 - Check "Enable DNS resolution" (should be enabled by default)
 - Check "Enable DNS hostnames"
- Click "Save"

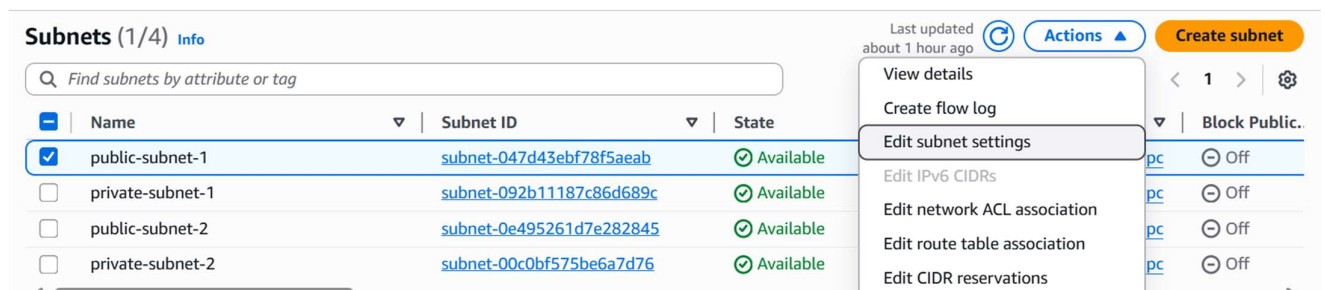


3. Enable Auto Assign Public IP in 2 public subnets

Purpose: To ensure instances launched in public subnets automatically receive public IP addresses for internet connectivity.

STEPS TO AUTO ASSIGN PUBLIC IPs IN PUBLIC SUBNETS:

- Navigate to **VPC Dashboard** → **Subnets**
- Select **public-subnet-1**
- Click "**Actions** → **Edit subnet settings**"



- Under "**Auto-assign IP settings**":
- Check "**Enable auto-assign public IPv4 address**"

Edit subnet settings [Info](#)

Subnet

Subnet ID
 subnet-047d43ebf78f5aeab

Name
 public-subnet-1

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)
Option disabled because no customer owned pools found.

Resource-based name (RBN) settings [Info](#)

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch [Info](#)

☐ Enable resource name DNS AAAA record on launch [Info](#)

Hostname type [Info](#)

☐ Resource name

☒ IP name

DNS64 settings

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

☐ Enable DNS64 [Info](#)

[Cancel](#) [Save](#)

- Click "Save"
- Repeat for **PublicSubnet2**

✓ You have successfully changed subnet settings:

- Enable auto-assign public IPv4 address

Subnets (1/4) [Info](#)

Last updated
less than a minute ago



[Actions](#) ▾

[Create subnet](#)

 Find subnets by attribute or tag

< 1 > 

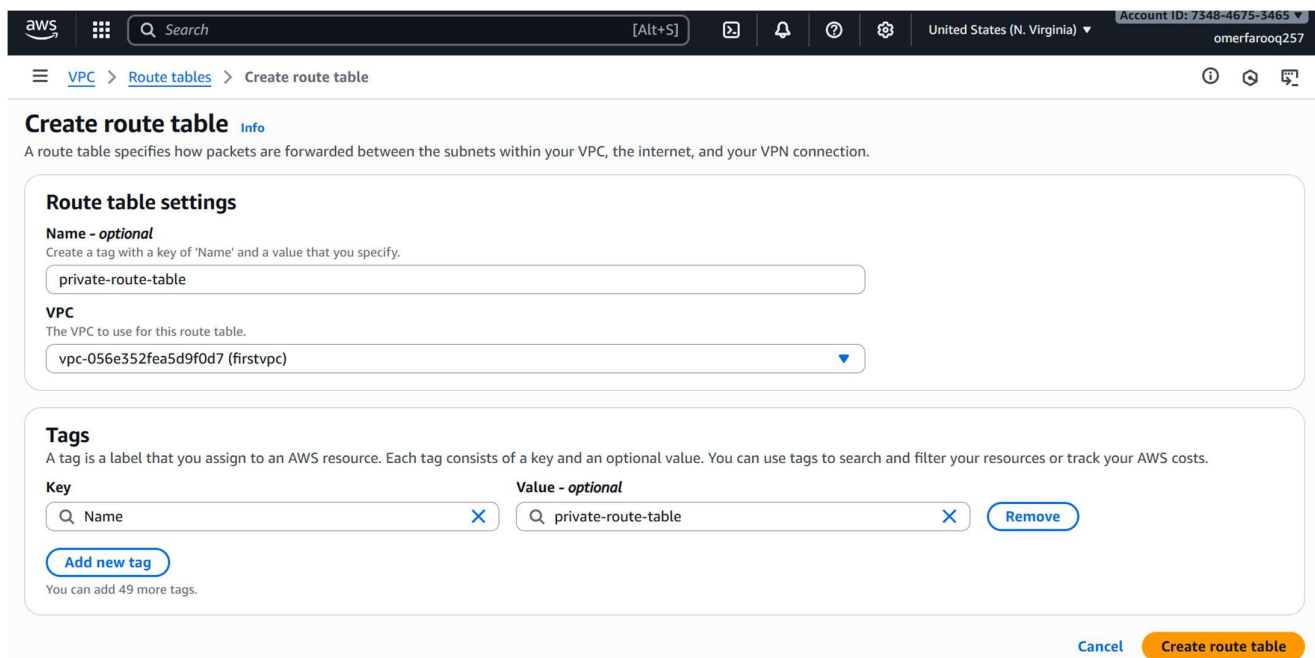
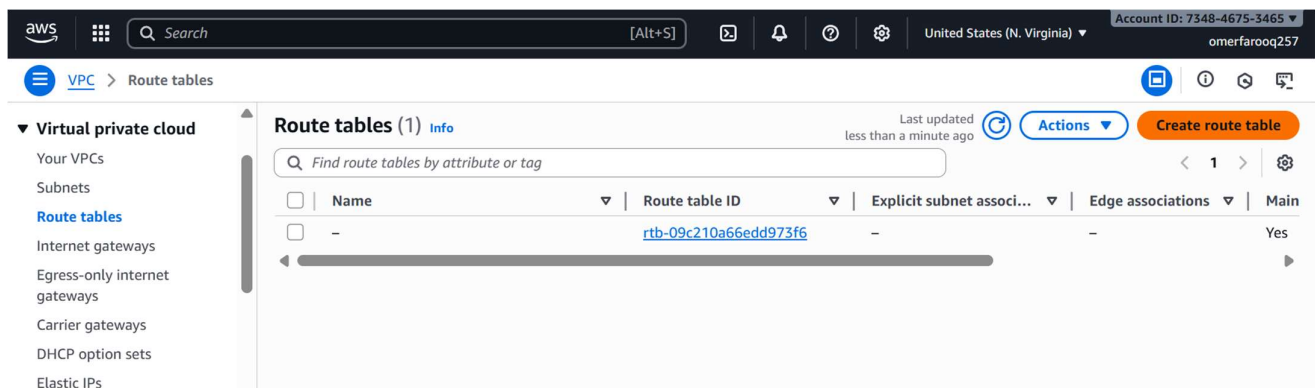
	Name	Subnet ID	State	VPC
☒	public-subnet-1	subnet-047d43ebf78f5aeab	✓ Available	vpc-056e352fea5d9f0d7 firstvpc

4. Add 2 private subnets in the private route table

Purpose: To control traffic routing for private subnets without direct internet access.

STEPS TO CREATE A ROUTE TABLE FOR PRIVATE SUBNETS:

- Navigate to **VPC Dashboard** → **Route Tables**
- Click "**Create route table**"
- Name: **private-route-table**
- VPC: Select **firstvpc**
- Click "**Create route table**"



- Verify the newly created Route Table.
- The details of the route table can be observed in the image below.
- In the Explicit subnet associations, there's nothing as of now. In the next step, we will associate the subnets with the route table.

Route table rtb-073a81f98fa842674 | private-route-table was created successfully.



rtb-073a81f98fa842674 / private-route-table

Actions ▾

Details [Info](#)

Route table ID

rtb-073a81f98fa842674

VPC

vpc-056e352fea5d9f0d7 |
firstvpc

Main

No

Owner ID

734846753465

Explicit subnet associations

–

Edge associations

–

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Both ▾

Edit routes

Filter routes

< 1 >

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/24	local	Active	No	Create Route Table

- Select the **private-route-table**
- Go to the "Subnet associations" tab
- Click "Edit subnet associations"
- Select both **private-subnet-1** and **private-subnet-2**
- Click "Save associations"

Route tables (1/2) [Info](#)

Last updated
1 minute ago



Actions ▾

Create route table

Find route tables by attribute or tag

< 1 >

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main
☐	–	rtb-09c210a66edd973f6	–	–	Yes
☒	private-route-table	rtb-073a81f98fa842674	–	–	No

rtb-073a81f98fa842674 / private-route-table



Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Details

Route table ID

rtb-073a81f98fa842674

VPC

vpc-056e352fea5d9f0d7 |
firstvpc

Main

No

Owner ID

734846753465

Explicit subnet associations

–

Edge associations

–

rtb-073a81f98fa842674 / private-route-table



Subnets without explicit associations (4)

[Edit subnet associations](#)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

< 1 >

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
public-subnet-1	subnet-047d43ebf78f5aeab	192.168.0.0/26	–
private-subnet-1	subnet-092b11187c86d689c	192.168.0.128/26	–
public-subnet-2	subnet-0e495261d7e282845	192.168.0.64/26	–
private-subnet-2	subnet-00c0bf575be6a7d76	192.168.0.192/26	–

[Alt+S]

VPC > Route tables > rtb-073a81f98fa842674 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

< 1 >

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	public-subnet-1	subnet-047d43ebf78f5aeab	192.168.0.0/26	–	Main (rtb-09c210a66edd973f6)
☒	private-subnet-1	subnet-092b11187c86d689c	192.168.0.128/26	–	Main (rtb-09c210a66edd973f6)
☐	public-subnet-2	subnet-0e495261d7e282845	192.168.0.64/26	–	Main (rtb-09c210a66edd973f6)
☒	private-subnet-2	subnet-00c0bf575be6a7d76	192.168.0.192/26	–	Main (rtb-09c210a66edd973f6)

Selected subnets

subnet-092b11187c86d689c / private-subnet-1

subnet-00c0bf575be6a7d76 / private-subnet-2

[Cancel](#) [Save associations](#)

rtb-073a81f98fa842674 / private-route-table

[Actions](#)Details [Info](#)

Route table ID
 rtb-073a81f98fa842674

VPC
[vpc-056e352fea5d9f0d7](#) | firstvpc

Main
☒ No

Owner ID
 734846753465

Explicit subnet associations
2 subnets
[subnet-092b11187c86d689c / private-subnet-1](#)
[subnet-00c0bf575be6a7d76 / private-subnet-2](#)

[Routes](#) | [Subnet associations](#) | [Edge associations](#) | [Route propagation](#) | [Tags](#)

Routes (1)

[Both](#) [Edit routes](#)

< 1 >

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/24	local	Active	No	Create Route Table

- Check the route table shows 2 explicit subnet associations
- Verify only local route exists (192.168.0.0/24 -> local)

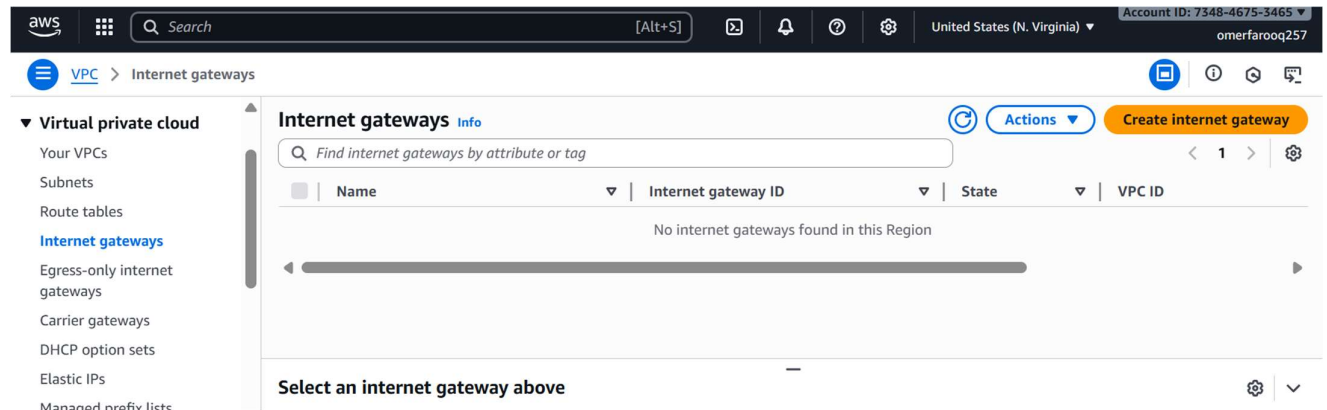
5. Add 2 public subnets in the public route table

Purpose: To enable internet connectivity for public-facing resources.

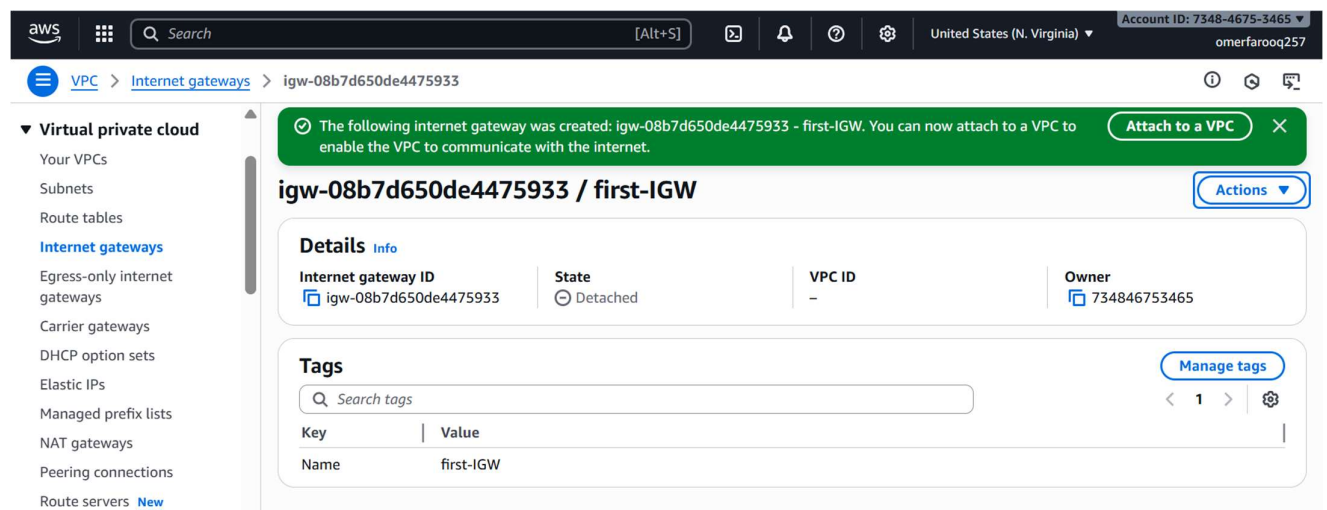
STEPS TO CREATE AN INTERNET GATEWAY AND A ROUTE TABLE FOR PUBLIC SUBNETS AND THEN ASSOCIATE PUBLIC SUBNETS TO THE PUBLIC ROUTE TABLE.

Create an Internet Gateway:

- Navigate to **VPC → Internet Gateways**
- Click "**Create internet gateway**"



- Name: **first-IGW**
- Click "**Create internet gateway**"



- Select the **IGW > Actions > "Attach to VPC"**
- Select **firstvpc**
- Click "**Attach internet gateway**"

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 7348-4675-3465

omerfarooq257

VPC > Internet gateways > Attach to VPC (igw-08b7d650de4475933)

Attach to VPC (igw-08b7d650de4475933)

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

vpc-056e352fea5d9f0d7 - firstvpc
AWS Command Line Interface Command

Cancel Attach internet gateway

Internet gateway igw-08b7d650de4475933 successfully attached to vpc-056e352fea5d9f0d7

igw-08b7d650de4475933 / first-IGW

Actions

Details

Internet gateway ID
igw-08b7d650de4475933

State
Attached

VPC ID
vpc-056e352fea5d9f0d7
firstvpc

Owner
734846753465

Tags

Manage tags

< 1 >

Key	Value
Name	first-IGW

Create Public Route Table:

- VPC → Route Tables → Create route table
- Name: **public-route-table**
- VPC: Select **firstvpc**
- Click "Create route table"

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
public-route-table

VPC
The VPC to use for this route table.
vpc-056e352fea5d9f0d7 (firstvpc)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Name

Value - optional
public-route-table

Remove

Add new tag

You can add 49 more tags.

Cancel

Create route table

Edit Subnet Associations:

- Select **public-route-table**
- **Subnet associations** → **Edit subnet associations**
- Select **public-subnet-1** and **public-subnet-2**
- Click "**Save associations**"

VPC > Route tables > rtb-0cac9ca196b250b79 > Edit subnet associations

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (2/4)

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	public-subnet-1	subnet-047d43ebf78f5aeab	192.168.0.0/26	–	Main (rtb-09c210a66edd973f6)
☐	private-subnet-1	subnet-092b11187c86d689c	192.168.0.128/26	–	rtb-073a81f98fa842674 / private-ro
☒	public-subnet-2	subnet-0e495261d7e282845	192.168.0.64/26	–	Main (rtb-09c210a66edd973f6)
☐	private-subnet-2	subnet-00c0bf575be6a7d76	192.168.0.192/26	–	rtb-073a81f98fa842674 / private-ro

Selected subnets

subnet-047d43ebf78f5aeab / public-subnet-1 X subnet-0e495261d7e282845 / public-subnet-2 X

Cancel Save associations

VPC > Route tables > rtb-0cac9ca196b250b79

rtb-0cac9ca196b250b79 / public-route-table

Details Info

Route table ID
rtb-0cac9ca196b250b79

VPC
vpc-056e352fea5d9f0d7 | firstvpc

Main
☐ No

Owner ID
734846753465

Explicit subnet associations
2 subnets

subnet-0e495261d7e282845 / public-subnet-2 X
subnet-047d43ebf78f5aeab / public-subnet-1

Routes Subnet associations Edge associations Route propagation Tags

Routes (1)

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/24	local	Active	No	Create Route Table

Both Edit routes

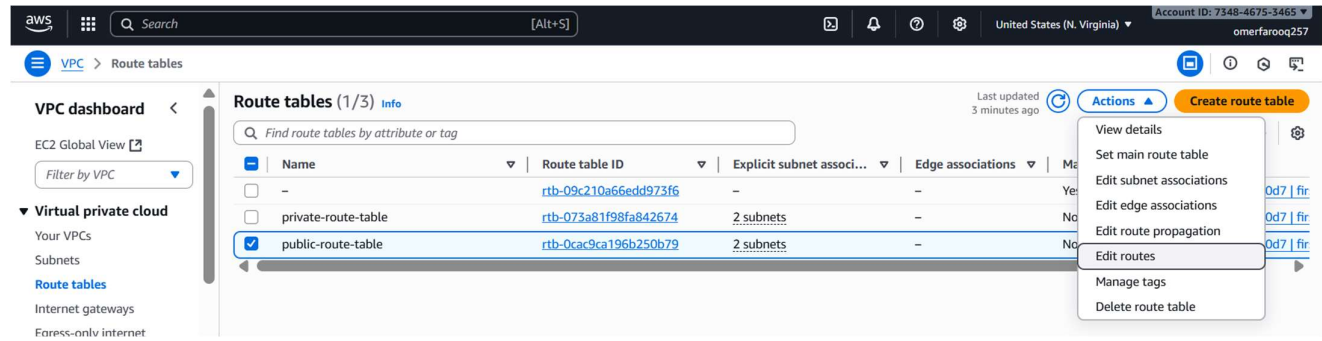
- Confirm 2 explicit subnet associations for public subnets

6. The public route table will have the routes to the internet and local

Purpose: To enable instances in public subnets to communicate with the internet.

STEPS TO EDIT/ASSIGN ROUTES TO PUBLIC ROUTE TABLE:

- Navigate to **VPC → Route Tables**
- Select **public-route-table**
- Go to **"Actions"** tab
- Click **"Edit routes"**



The screenshot shows the AWS Management Console interface for the 'Route tables' page. The left sidebar contains the 'VPC dashboard' and 'Virtual private cloud' sections. The main content area displays a table of route tables. The 'public-route-table' is selected, and the 'Actions' dropdown menu is open, showing the 'Edit routes' option.

Name	Route table ID	Explicit subnet associ...	Edge associations	Ma
-	rtb-09c210a66edd973f6	-	-	Yes
private-route-table	rtb-073a81f98fa842674	2 subnets	-	No
public-route-table	rtb-0cac9ca196b250b79	2 subnets	-	No

The 'Edit routes' page shows a table with the following columns: Destination, Target, Status, Propagated, and Route Origin. The 'Destination' is 192.168.0.0/24, the 'Target' is local, the 'Status' is Active, the 'Propagated' status is No, and the 'Route Origin' is CreateRouteTable. There is an 'Add route' button and 'Cancel', 'Preview', and 'Save changes' buttons at the bottom.

- Click **"Add route"**
- Destination: **0.0.0.0/0**
- Target: Select **"Internet Gateway" > first-IGW**
- Click **"Save changes"**

☰ VPC > Route tables > rtb-0cac9ca196b250b79 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
192.168.0.0/24	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Updated routes for rtb-0cac9ca196b250b79 / public-route-table successfully

[Details](#)

rtb-0cac9ca196b250b79 / public-route-table

Details

Route table ID
 rtb-0cac9ca196b250b79

VPC
vpc-056e352fea5d9f0d7 | firstvpc

Main
 No

Owner ID
 734846753465

Explicit subnet associations
2 subnets

Edge associations
-

[Routes](#)
[Subnet associations](#)
[Edge associations](#)
[Route propagation](#)
[Tags](#)

Routes (2)

Both
Edit routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-08b7d650de4475...	Active	No	Create Route
192.168.0.0/24	local	Active	No	Create Route Table

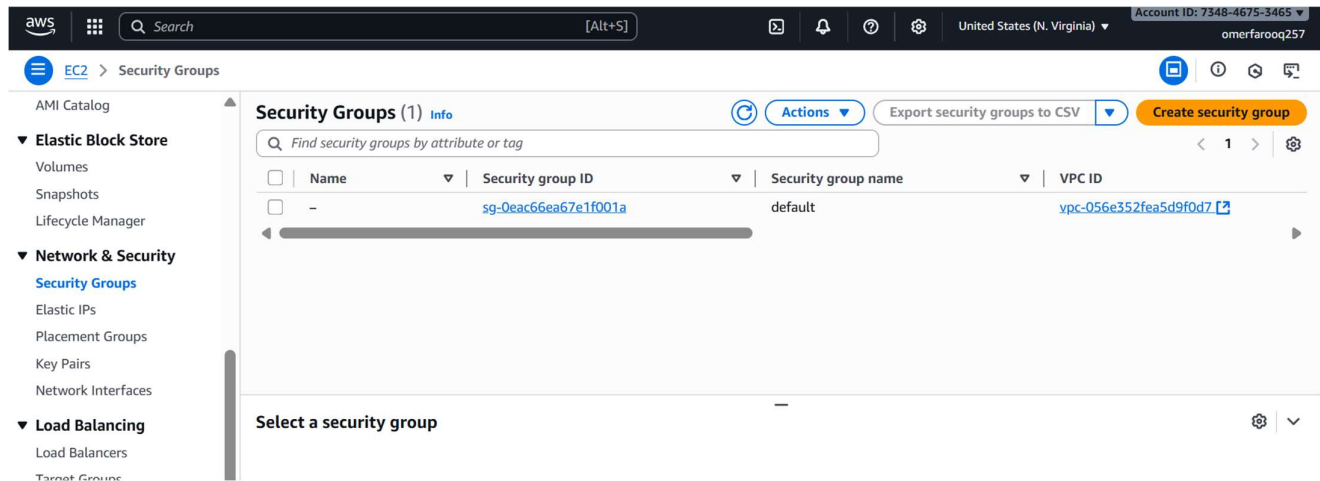
- Verify both local and internet routes exist
- Status should be "active" for both routes

7. Create EC2 in a public subnet with t2micro and install PHP

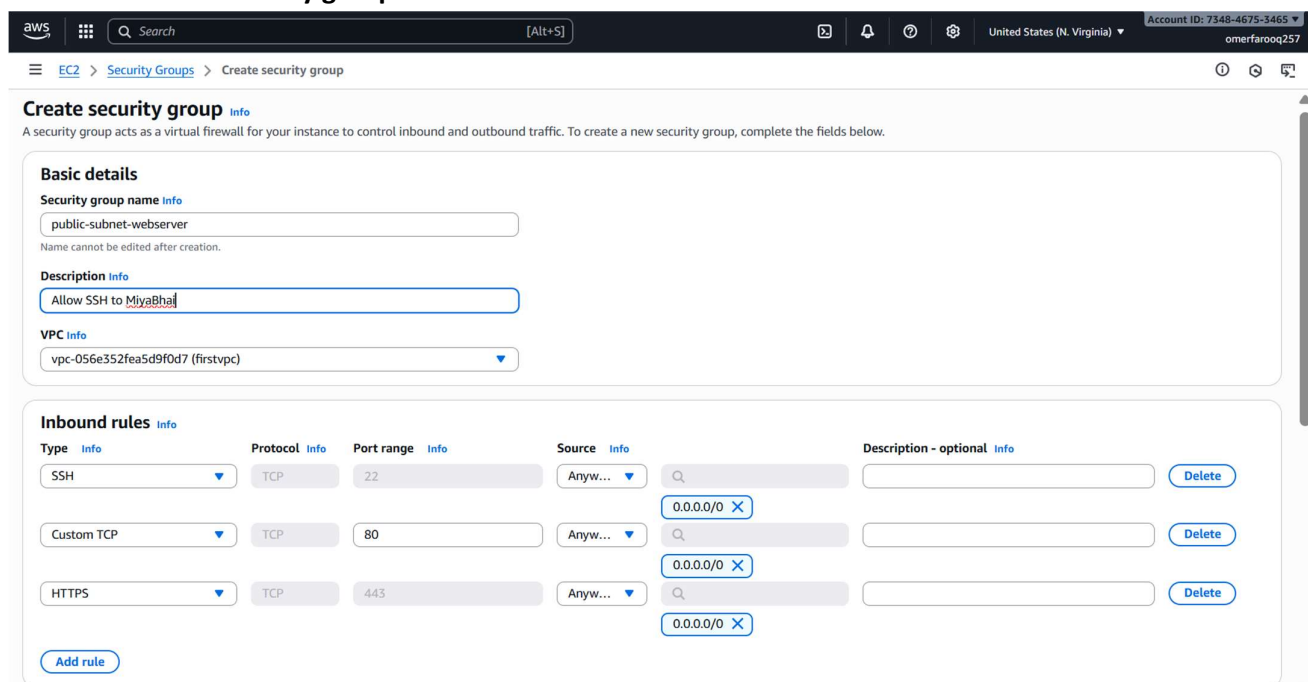
Purpose: To host public-facing web applications with PHP support

STEPS TO LAUNCH AND EC2 INSTANCE WITH A CUSTOM SECURITY GROUP, INSTALL PHP, AND RUN A STATIC WEBSITE USING HTTPD.

- Navigate to **EC2 → Security Groups**
- Click "**Create security group**"



- Name: **public-subnet-webserver**
- Description: <Custom description>
- VPC: Select **firstvpc**
- Inbound rules:
 - **SSH: TCP, Port 22, Source: 0.0.0.0/0 (or your IP for security)**
 - **HTTP: TCP, Port 80, Source: 0.0.0.0/0**
 - **HTTPS: TCP, Port 443, Source: 0.0.0.0/0**
- Click "**Create security group**"



Security group (sg-0dc65848d27986cba | public-subnet-webserver) was created successfully
Details

sg-0dc65848d27986cba - public-subnet-webserver

Actions

Details

Security group name
public-subnet-webserver

Security group ID
sg-0dc65848d27986cba

Description
Allow SSH to MiyaBhai

VPC ID
vpc-056e352fea5d9f0d7

Owner
734846753465

Inbound rules count
3 Permission entries

Outbound rules count
1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (3)



Manage tags

Edit inbound rules

Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-00b5dd91f4c887f50	IPv4	SSH	TCP	22
☐	-	sgr-07ce88abdf1731c09	IPv4	HTTP	TCP	80
☐	-	sgr-088d044af054dcc6c	IPv4	HTTPS	TCP	443

Create an EC2 Instance with t2.micro:

EC2 > Instances > Launch an instance

Instance type

t2.micro

Free tier eligible

Key pair (login)

ec2

Create new key pair

Network settings

VPC: vpc-056e352fea5d9f0d7 | firstvpc

Subnet: subnet-047d45ebf78f5aeb | public-subnet-1

Auto-assign public IP: Enable

Firewall (security groups)

Create security group

Select existing security group

Common security groups

public-subnet-webserver

default

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.9.2

Virtual server type (instance type): t2.micro

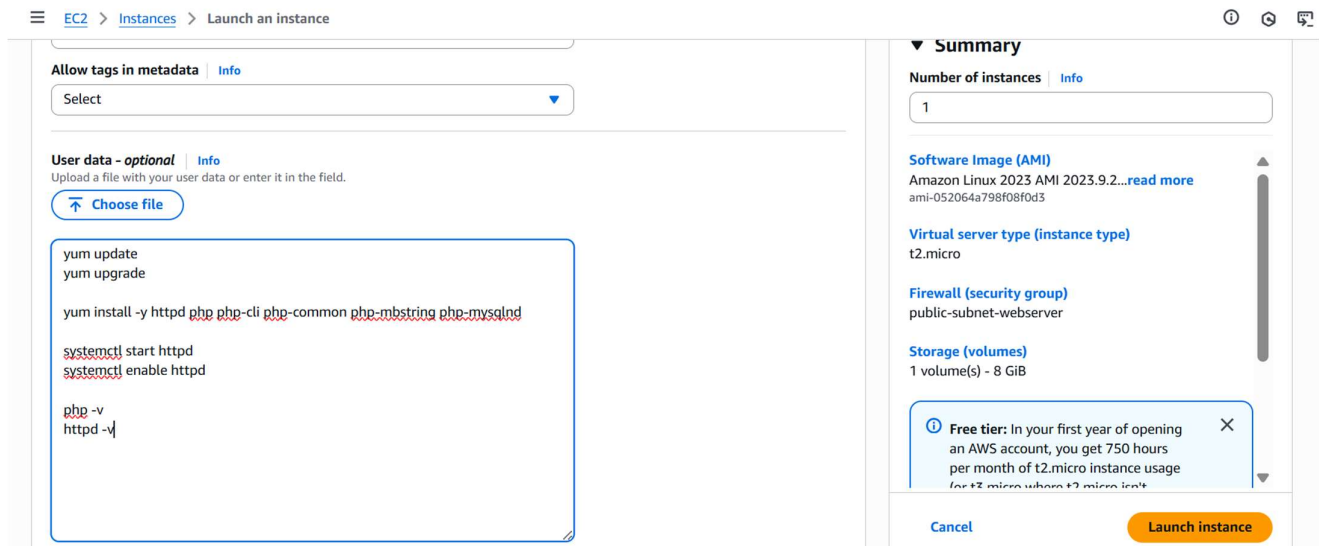
Firewall (security group): public-subnet-webserver

Storage (volumes): 1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

Launch instance

- Navigate to **EC2 Dashboard**
- Click "**Launch instances**"
- **Name:** PublicWebServer
- **AMI:** Amazon Linux 2023 AMI or Ubuntu 22.04 LTS
- Instance type: **t2.micro**
- **Key pair:** Select or create a key pair
- Network settings:
 - VPC: **firstvpc**
 - Subnet: **public-subnet-1**
 - Auto-assign public IP: **Enable**
 - Security group: Select "**public-server-subnet**"
- Click "**Launch instance**"

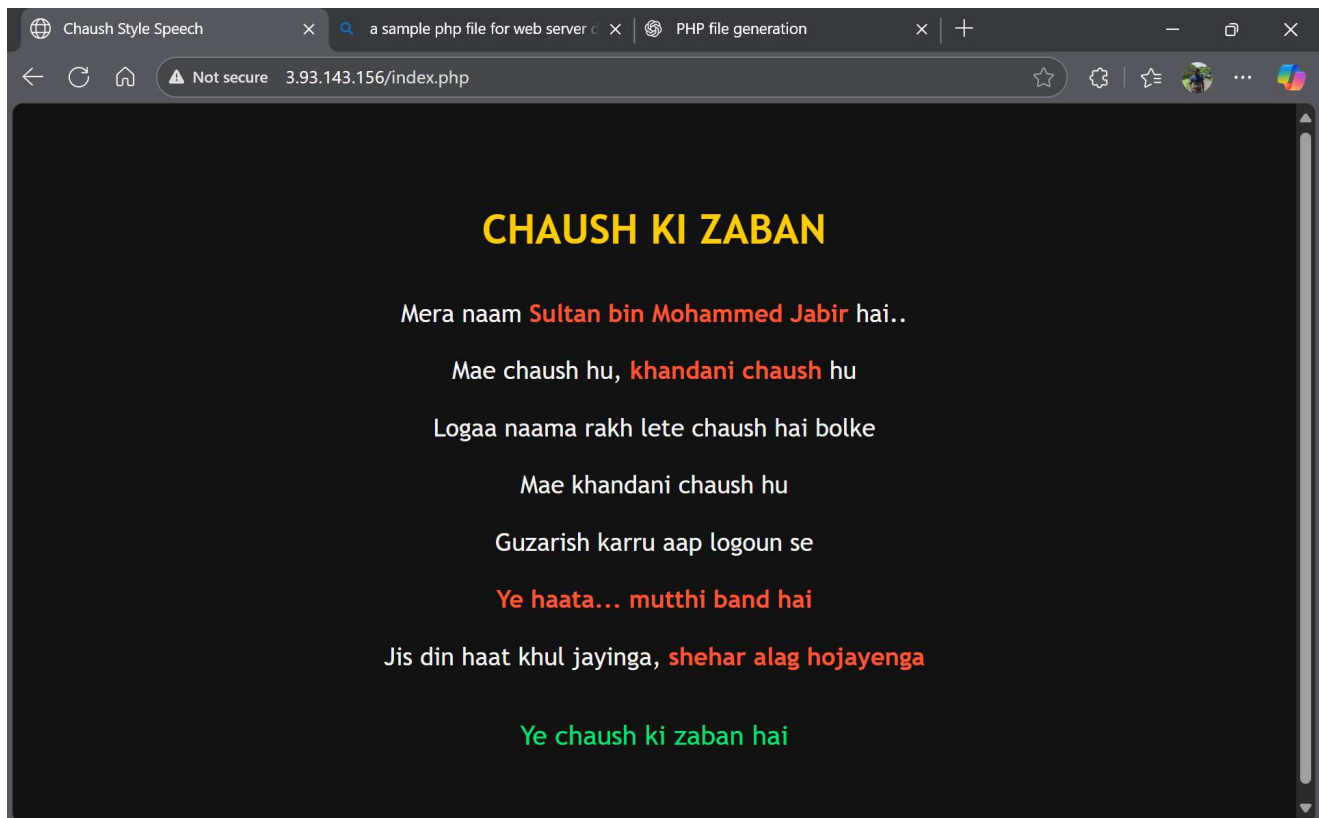


Connect to the instance:

- Connect to the instance **using SSH**.
- Check if **PHP** and **HTTPD** are installed properly. If not, install manually.
- Create a sample file “**index.php**” in “**/var/www/html/**”
- Add any content and save the file.

Verification:

- Access `http://<public-ip>/index.php` in browser



8. Configure the NAT gateway in the public subnet and connect to the private Instance

Purpose: To allow private instances to access the internet for updates while remaining private.

STEPS TO GET AN ELASTIC IP AND CONFIGURE A NAT GATEWAY IN THE PUBLIC SUBNET. THEN CREATE A ROUTE IN THE PRIVATE ROUTE TABLE.

Allocate Elastic IP:

- Navigate to **VPC Dashboard > Elastic IPs**
- Click "**Allocate Elastic IP address**"
- **Network Border Group: Default**
- Click "**Allocate**"
- Note the Allocation ID

The image shows two screenshots from the AWS VPC console. The top screenshot displays the 'Elastic IP addresses' page, which is currently empty, showing a message 'No Elastic IP addresses found in this Region'. The bottom screenshot shows the 'Allocate Elastic IP address' configuration page. In this page, the 'Public IPv4 address pool' is set to 'Amazon's pool of IPv4 addresses', and the 'Network border group' is set to 'us-east-1'. There are also sections for 'Global static IP addresses' and 'Tags - optional'.

Elastic IP addresses

Find elastic IP addresses by attribute or tag

Name	Allocated IPv4 address	Type	Allocation ID
No Elastic IP addresses found in this Region			

Select an elastic IP address

View IP address usage and recommendations to release unused IPs with [Public IP insights](#)

Allocate Elastic IP address

Elastic IP address settings

Public IPv4 address pool

- ☒ Amazon's pool of IPv4 addresses
 - Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) [Learn more](#)
 - Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) [Learn more](#)
 - Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group

us-east-1

Global static IP addresses

AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. [Learn more](#)

[Create accelerator](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tag

[Cancel](#) [Allocate](#)

✓ Elastic IP address allocated successfully.
Elastic IP address 54.204.109.30
Associate this Elastic IP address
✕

Elastic IP addresses (1) [Info](#)

Find elastic IP addresses by attribute or tag

Public IPv4 address : 54.204.109.30 ✕ Clear filters

☐	Name	Allocated IPv4 addr...	Type	Allocation ID
☐	-	54.204.109.30	Public IP	eipalloc-03e92c2e33ff340bd

Create NAT Gateway:

- Navigate to **VPC Dashboard > NAT Gateways**
- Click **"Create NAT gateway"**
- Name: **First-NAT-Gateway**
- Subnet: Select **public-subnet-1**

aws
Search
[Alt+S]
United States (N. Virginia)
Account ID: 7348-4675-3465
omerfarooq257

VPC > NAT gateways > Create NAT gateway

Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

NAT gateway settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Subnet
Select a subnet in which to create the NAT gateway.

subnet-04663953814f438f5 (Private-Subnet-2)
use1-az1 (us-east-1d)

subnet-0e99d35a7b9d41f8b (Public-Subnet-1)
use1-az2 (us-east-1a)

subnet-09e6c23c0cf0f3a6e (Private-Subnet-1)
use1-az6 (us-east-1c)

subnet-0ae6123aea408db68 (Public-Subnet-2)
use1-az4 (us-east-1b)

Allocate Elastic IP

- Elastic IP allocation ID: **Select the allocated EIP**
- Click **"Create NAT gateway"**

aws
Search
[Alt+S]
United States (N. Virginia)
Account ID: 7348-4675-3465
omerfarooq257

VPC > NAT gateways > Create NAT gateway

Select a subnet in which to create the NAT gateway.

subnet-0e99d35a7b9d41f8b (Public-Subnet-1)

Connectivity type
Select a connectivity type for the NAT gateway.

☒ Public
☐ Private

Elastic IP allocation ID [Info](#)
Assign an Elastic IP address to the NAT gateway.

Select an Elastic IP

eipalloc-03e92c2e33ff340bd
54.204.109.30 us-east-1

Allocate Elastic IP

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Remove

Add new tag

You can add 49 more tags.

Cancel
Create NAT gateway

Actions ▼

NAT gatev

—

```
i amom@Omer MINGW64 ~/Downloads ~\Downloads
$ ssh -i "vpc.pem" ec2-user@ec2-3-88-230-238.compute-1.amazonaws.com

#_
~\_ #####_ Amazon Linux 2023
~~ \#####\
~~ \###|
~~ \#/ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~~
~~.-.-
~~ \_/ \_/
~~ _/m/' '->

Last login: Fri Oct 17 22:00:44 2025 from 49.206.52.62
[ec2-user@ip-192-168-1-5 ~]$ sudo su
[root@ip-192-168-1-5 ec2-user]# vi key.pem
[root@ip-192-168-1-5 ec2-user]# chmod 400 key.pem
[root@ip-192-168-1-5 ec2-user]# ssh -i key.pem ec2-user@192.168.1.140

#_
~\_ #####_ Amazon Linux 2023
~~ \#####\
~~ \###|
~~ \#/ https://aws.amazon.com/linux/amazon-linux-2023
~~ V~' '->
~~~~
~~.-.-
~~ \_/ \_/
~~ _/m/' '->

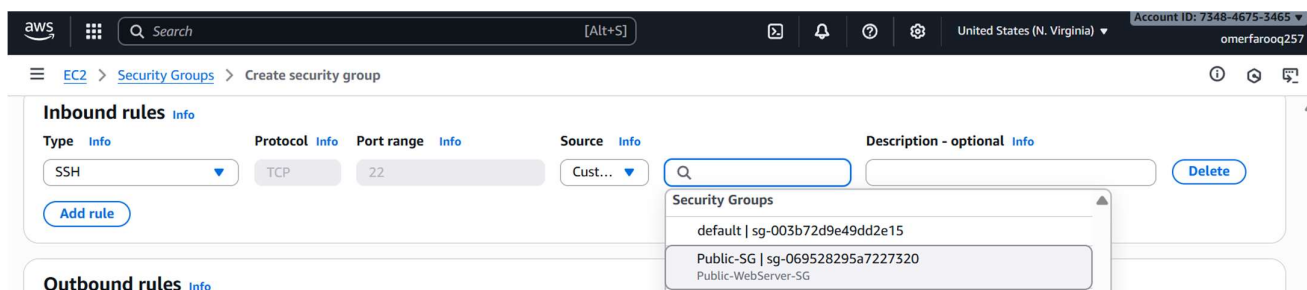
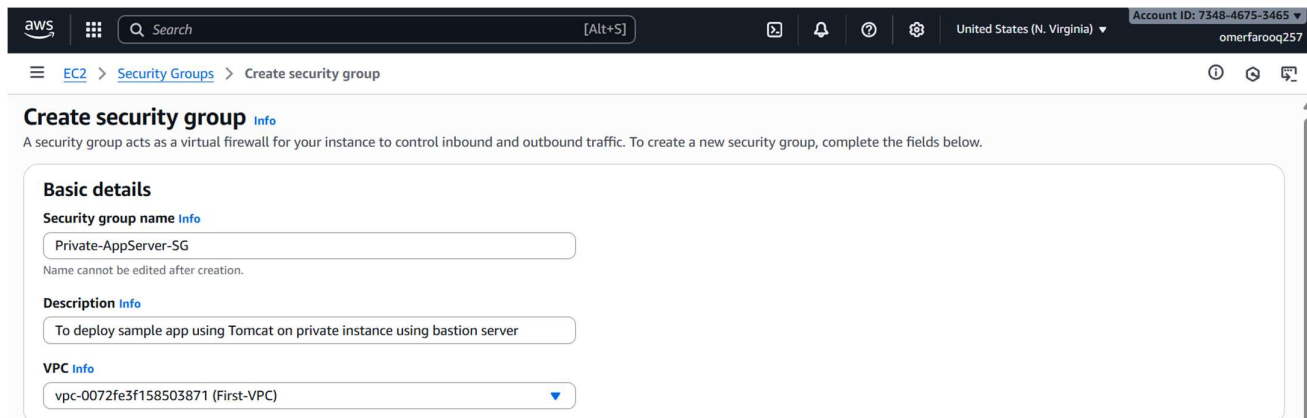
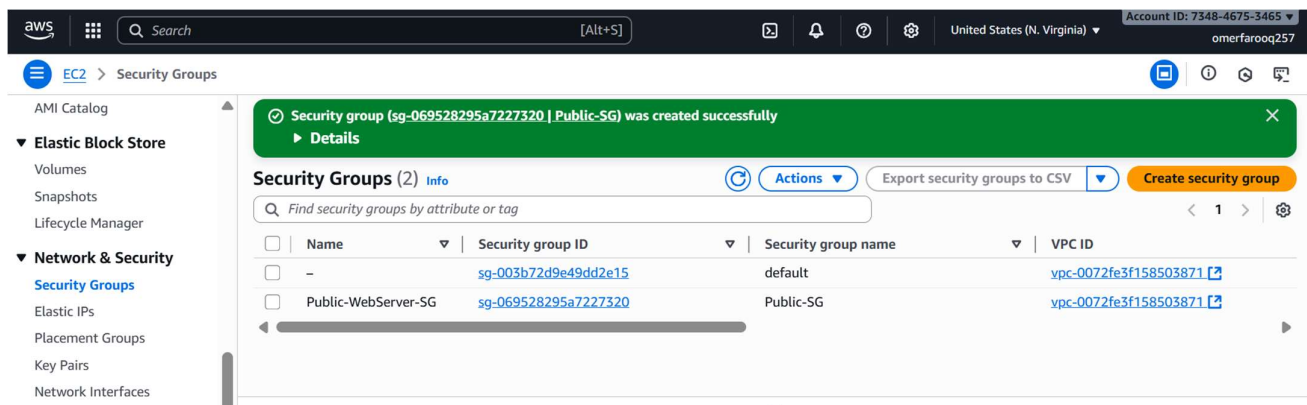
Last login: Fri Oct 17 22:02:33 2025 from 192.168.1.5
[ec2-user@ip-192-168-1-140 ~]$ sudo su
[root@ip-192-168-1-140 ec2-user]#
```

9. Install Apache Tomcat in a private EC2 and deploy a sample app.

Purpose: To host Java applications in a secure, private environment

STEPS TO INSTALL APACHE TOMCAT IN A PRIVATE EC2 INSTANCE AND DEPLOY A SAMPLE APPLICATION:

- **EC2 Dashboard > Security Groups > Create security group**
- Name: **Private-SG**
- VPC: **firstvpc**
- Inbound rules:
 - **SSH: TCP, Port 22, Source: WebServerSG security group**
 - **Custom TCP: Port 8080, Source: 192.168.0.0/24**
- Click **"Create security group"**



Steps to launch an EC2 Instance in a Private Subnet:

- EC2 Dashboard > Launch instances
- **Name:** <Instance_Name>
- **AMI:** Amazon Linux 2023 AMI
- Instance type: **t2.micro**
- **Key pair:** Same as public instance
- Network settings:
 - VPC: **firstvpc**
 - Subnet: **private-subnet-1**
 - Auto-assign public IP: **Disable**
 - Security group: **Private-SG**
- Click "**Launch instance**"

Instance summary for i-07245e13d74c340d8 (AppServer-Pvt) [Info](#)

Updated less than a minute ago

Instance ID i-07245e13d74c340d8	Public IPv4 address -	Private IPv4 addresses 192.168.0.152
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-192-168-0-152.ec2.internal	Private IP DNS name (IPv4 only) ip-192-168-0-152.ec2.internal	
Answer private resource DNS name -	Instance type t2.micro	Elastic IP addresses -
Auto-assigned IP address -	VPC ID vpc-0072fe3f158503871 (First-VPC)	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
IAM Role -	Subnet ID subnet-09e6c23c0cf0f3a6e (Private-Subnet-1)	Auto Scaling Group name -
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:734846753465:instance/i-07245e13d74c340d8	Managed false
Operator -		

Connect to Private Instance via Bastion or Jump Server:

- To connect to the private instance, we first need to connect to the public instance, and then, from the public (Bastion/Jump) server, we connect to the private instance.
- The key that we use to connect to the public instance is used to connect to the private instance. We can either copy the key, create a new file and paste it into the bastion server, or copy the "key.pem" file directly to the bastion server.

```
# scp -i your-key.pem file.pem ec2-user@<public-ip>:/path/to/the/directory
```

```
# scp -l vpc.pem vpc.pem ec2-user@3.243.86.54:/home/ec2-user
```

- Connect to the Public Instance/Bastion/Jump Server

```
# ssh -i your-key.pem ec2-user@<public-ip>
```

- The newly created key.pem file requires 400 permission to connect to the private instance.

```
# chmod vpc.pem -> from the public instance.
```

- Connect to the private instance.

```
# ssh -i your-key.pem ec2-user@<private-ip>
```

```
#ssh -l vpc.pem ec2-user@192.168.0.150
```



```

root@ip-192-168-0-152/tmp
[root@ip-192-168-0-152 ec2-user]# cd /tmp
[root@ip-192-168-0-152 tmp]# wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.11/bin/apache-tomcat-11.0.11.tar.gz
--2025-10-05 13:27:03-- https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.11/bin/apache-tomcat-11.0.11.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14137461 (13M) [application/x-gzip]
Saving to: 'apache-tomcat-11.0.11.tar.gz'

apache-tomcat-11.0.11.tar. 100%[=====>] 13.48M --.-KB/s in 0.09s

2025-10-05 13:27:03 (143 MB/s) - 'apache-tomcat-11.0.11.tar.gz' saved [14137461/14137461]

[root@ip-192-168-0-152 tmp]# ls
apache-tomcat-11.0.11.tar.gz
hsperfdata_root
systemd-private-77c93b6e1d4f4b8887a5a10c621e6192-chronyd.service-uAImk2
systemd-private-77c93b6e1d4f4b8887a5a10c621e6192-dbus-broker.service-1lVjvw
systemd-private-77c93b6e1d4f4b8887a5a10c621e6192-policy-routes@enX0.service-yEm5Ej
systemd-private-77c93b6e1d4f4b8887a5a10c621e6192-systemd-logind.service-tWN6Ru
systemd-private-77c93b6e1d4f4b8887a5a10c621e6192-systemd-resolved.service-BSBuyum
[root@ip-192-168-0-152 tmp]# tar xvf apache-tomcat-11.0.11.tar.gz -C /opt/tomcat --strip-components=1
apache-tomcat-11.0.11/conf/
apache-tomcat-11.0.11/conf/catalina.properties
apache-tomcat-11.0.11/conf/context.xml
apache-tomcat-11.0.11/conf/jaspic-providers.xml
apache-tomcat-11.0.11/conf/jaspic-providers.xsd

```

Configure Tomcat as a Service

- Create a “**tomcat.service**” file at “**/etc/systemd/system/**”

```

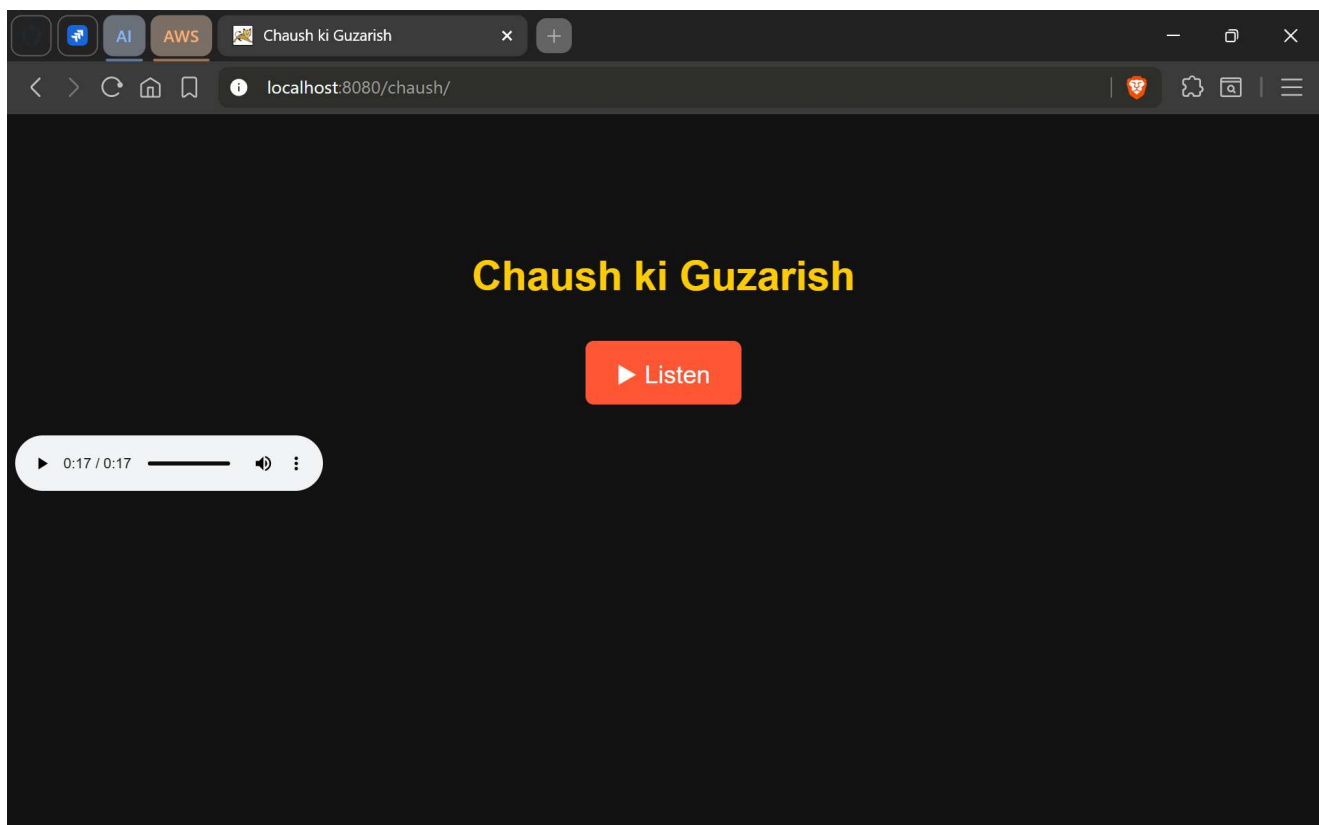
root@ip-192-168-0-152/tmp
[root@ip-192-168-0-152 tmp]# vi /etc/systemd/system/tomcat.service
[root@ip-192-168-0-152 tmp]# chmod 644 /etc/systemd/system/tomcat.service
[root@ip-192-168-0-152 tmp]# ls -la /etc/systemd/system/tomcat.service
-rw-r--r--. 1 root root 644 Oct 5 13:29 /etc/systemd/system/tomcat.service
[root@ip-192-168-0-152 tmp]# systemctl daemon-reload
[root@ip-192-168-0-152 tmp]# systemctl start tomcat.service
Job for tomcat.service failed because the control process exited with error code.
See "systemctl status tomcat.service" and "journalctl -xeu tomcat.service" for details.
[root@ip-192-168-0-152 tmp]# chown -R tomcat:tomcat /opt/tomcat
[root@ip-192-168-0-152 tmp]# chmod -R 750 /opt/tomcat
[root@ip-192-168-0-152 tmp]# systemctl daemon-reload
[root@ip-192-168-0-152 tmp]# systemctl start tomcat.service
[root@ip-192-168-0-152 tmp]# systemctl status tomcat
● tomcat.service - Apache Tomcat 11 Web Application Container
   Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: disabled)
   Active: active (running) since Sun 2025-10-05 13:35:59 UTC; 48s ago
 Main PID: 27042 (java)
    Tasks: 30 (limit: 1106)
   Memory: 128.1M
      CPU: 3.514s
  CGroup: /system.slice/tomcat.service
          └─27042 /usr/lib/jvm/java-21-amazon-corretto.x86_64/bin/java -Djava.util.logging.config.file>

Oct 05 13:35:59 ip-192-168-0-152.ec2.internal systemd[1]: Starting tomcat.service - Apache Tomcat 11 Web>
Oct 05 13:35:59 ip-192-168-0-152.ec2.internal startup.sh[27035]: Tomcat started.
Oct 05 13:35:59 ip-192-168-0-152.ec2.internal systemd[1]: Started tomcat.service - Apache Tomcat 11 Web>
lines 1-13/13 (END)

```

- Check if the Tomcat service is running.
Active: active (running)


```
ec2-user@ip-192-168-0-26:~  
iamom@Omer MINGW64 ~/Downloads  
$ ssh -i vpc -L 8080:192.168.0.152:8080 ec2-user@44.203.3.8  
Warning: Identity file vpc not accessible: No such file or directory.  
ec2-user@44.203.3.8: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).  
iamom@Omer MINGW64 ~/Downloads  
$ ssh -i vpc.pem -L 8080:192.168.0.152:8080 ec2-user@44.203.3.8  
#_   
'\ #####_ Amazon Linux 2023  
~~ \#####\  
~~ \###|  
~~ \#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~ V~', '->  
~~~  
~~~  
~~~  
_m/,'-/->  
Last login: Sun Oct 5 13:38:47 2025 from 152.59.201.50  
[ec2-user@ip-192-168-0-26 ~]$
```



Verification:

To check if the Deployed app is running properly, log in to the Bastion Server and check using:

```
# curl http://<private-instance-ip>:8080/chaush/
```

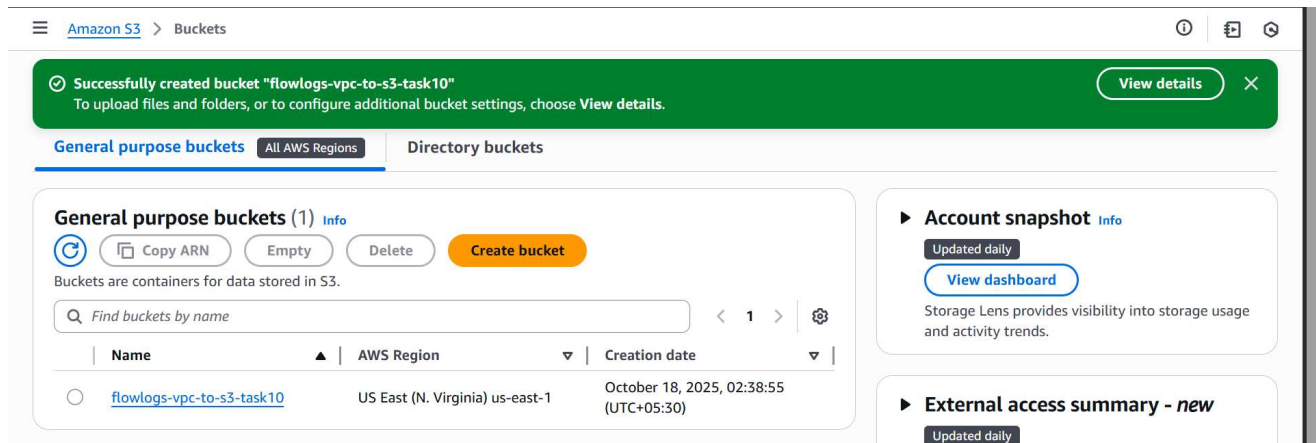
10. Configure VPC flow logs and store the logs in S3 and CloudWatch.

Purpose: To capture network traffic information for security analysis, troubleshooting, and compliance

STEPS TO CONFIGURE VPC FLOW LOGS AND STORE THE LOGS IN S3 AND CLOUDWATCH:

Step 1: Create an S3 Bucket for Flow Logs

- Navigate to **S3 Console**
- Click "**Create bucket**"
- Bucket name: flowlogs-vpc-s3
- **Region: Same as your VPC**
- **Block all public access:** Enable
- **Bucket versioning:** Enable (optional)
- Click "**Create bucket**"



Step 2: Create an IAM Role for Flow Logs to S3

- Navigate to **IAM Console > Roles**
- Click "**Create role**"
- Trusted entity type: **AWS service**
- Use case: **VPC - Flow Logs**
- Click "**Next**"
- Create inline policy:
 - Service: **S3**
 - Actions: **PutObject, GetBucketLocation**
 - Resources: **Specify your S3 bucket ARN**
- Role name: **VPCFlowLogsToS3Role**
- Click "**Create role**"

Identity and Access Management (IAM)

Search IAM

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VPCFlowLogsToS3Role [Info](#)

[Delete](#)

[Edit](#)

Summary

Creation date
October 18, 2025, 02:46 (UTC+05:30)

ARN
[arn:aws:iam::734846753465:role/VPCFlowLogsToS3Role](#)

Last activity
-

Maximum session duration
1 hour

[Permissions](#) | [Trust relationships](#) | [Tags](#) | [Last Accessed](#) | [Revoke sessions](#)

Permissions policies (1) [Info](#)

You can attach up to 10 managed policies.

[Simulate](#) [Remove](#) [Add permissions](#)

Filter by Type
All types

Search

☐	Policy name	Type	Attached entities
☐	VPCFlowLogsToS3Policy	Customer inline	0

Step 3: Create VPC Flow Log to S3

- Navigate to **VPC Dashboard**
- Select your **VPC** (Test-VPC)
- Go to the **"Flow logs"** tab
- Click **"Create flow log"**
- Name: **VPC-FlowLog-S3**
- Filter: **All**
- Maximum aggregation interval: **10 minutes (or 1 minute)**
- Destination: **Send to an Amazon S3 bucket**
- **S3 bucket ARN**: `arn:aws:s3:::vpc-flow-logs-<account-id>-<region>/vpc-flow-logs/`
- **Log record format**: AWS default format (or custom format)
- Tags: Add relevant tags
- Click **"Create flow log"**

[Amazon S3](#) > ... > [AWSLogs/](#) > [734846753465/](#) > [vpcflowlogs/](#) > [us-east-1/](#) > [2025/](#) > [10/](#) > 17/

17/ [Copy S3 URI](#)

[Objects](#) | [Properties](#)

Objects (1) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	734846753465_vpcflowlogs_us-east-1_fl-018ec7740ab84ca28_20251017T2140Z_5632219e.log.gz	gz	October 18, 2025, 03:14:12 (UTC+05:30)	358.0 B	Standard

VPC Flow log stored in an S3 bucket can be seen below

```
version account-id interface-id srcaddr dstaddr srcport dstport protocol packets bytes start end action log-status
2 734846753465 eni-035bc0a682b9cf08a 79.124.56.6 192.168.1.4 40011 57445 6 1 40 1760737370 1760737370 ACCEPT OK
2 734846753465 eni-035bc0a682b9cf08a 147.185.133.163 192.168.1.4 54003 340 6 1 44 1760737374 1760737374 ACCEPT OK
2 734846753465 eni-035bc0a682b9cf08a 176.65.148.234 192.168.1.4 46915 8728 6 1 40 1760737382 1760737386 ACCEPT OK
2 734846753465 eni-035bc0a682b9cf08a 167.94.145.26 192.168.1.4 4376 2404 6 1 60 1760737382 1760737386 ACCEPT OK
2 734846753465 eni-035bc0a682b9cf08a - - - - - 1760737413 1760737413 - NODATA
2 734846753465 eni-035bc0a682b9cf08a - - - - - 1760737413 1760737413 - SKIPDATA
2 734846753465 eni-035bc0a682b9cf08a 79.124.56.6 192.168.1.4 40005 57445 6 1 40 1760737392 1760737415 ACCEPT OK
2 734846753465 eni-035bc0a682b9cf08a 35.203.210.224 192.168.1.4 55250 1902 17 1 125 1760737392 1760737415 ACCEPT OK
2 734846753465 eni-035bc0a682b9cf08a 79.124.56.6 192.168.1.4 40030 57445 6 1 40 1760737387 1760737392 ACCEPT OK
2 734846753465 eni-035bc0a682b9cf08a 204.76.203.212 192.168.1.4 35692 80 6 1 52 1760737387 1760737392 ACCEPT OK
```